CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. (EC) Examination

May 2017

EC404.01/EC404 Embedded Systems

Date: 25.05.2017, Thursday **Time: 01:30 p.m. to 04:30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.

SECTION – I

Q - 1 (a) Discuss the following content of I2C bus: **[07]**

i) structure, ii) electrical interface, iii) format of an I2C data frame, iv) application interface.

(b) The Design metrics play very important role in Embedded System. Define it and explain four design metrics of embedded systems with relevant example. **[07]**

Q - 2 Answer the following questions (any three) **[21]**

1. Interface DC motor with LPC2148 ARM7 processor and write a program to rotate DC motor clock wise and anti clock wise using lpc2148 development board.
2. Explain Processor technology which used for designing Embedded Systems. Compare all three major category named
   i) General Purpose, ii) Single Purpose, iii) Application Specific.
3. What is RTOS? How does it differ from conventional OS? Describe four function and their activities provided by RTOS services.
4. With neat diagram, Interface LPC2148 with LCD, also write a C program to display message “WELCOME TO” on first line and “CHARUSAT” on second line.
5. With the help of example, Explain following RTOS scheduling

   i) Cooperative Scheduling of Ready Tasks using Precedence Constrains,

   ii) Preemptive scheduling.

SECTION – II

Q - 3 Describe the main phases of a design review. **[07]**

Q – 4 (a) Compare RICS and CISC processor. Why the RISC is more suitable for embedded system like network processor? **[07]**

(b) With help of suitable example, explain data processing instruction of ARM processor. **[07]**

OR

(b) Explain following ARM processor instruction in detail **[07]**

a) Base plus offset addressing mode instructions,

b) Multiple load store instruction

Q-5 (a) Briefly describe the differences between the waterfall and spiral development models. **[07]**

(b) Write program to count 00 to 99 on two 7-Segment Display which interface with LPC2148 ARM processor. **[07]**

OR

(b) Write a Technical note on Quality Assurance in Embedded System Design. **[07]**